From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

JEE(Main), 2014

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST –V
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	A	C
2.	C	C	B
3.	A	B	C
4.	D	D	B
5.	C	C	B
6.	A	B	A
7.	D	D	A
8.	B	D	B
9.	C	B	C
10.	B	A	D
11.	C	B	C
12.	C	C	B
13.	A	B	D
14.	B	C	B
15.	A	D	A
16.	C	D	C
17.	C	B	B
18.	B	C	C
19.	D	B	A
20.	A	B	A
21.	A	C	B
22.	B	B	B
23.	B	D	C
24.	C	D	B
25.	C	C	C
26.	D	A	A
27.	C	B	C
28.	C	C	D
29.	C	A	B
30.	D	A	C

Physics

PART – I

1. Viscosity of gases increases & that of liquid decreases.

2. $T - mg\cos\theta = \frac{mu^2}{\ell}$

3. $x = 2t^2$

$\Rightarrow v\frac{dv}{dx} = 4 \Rightarrow v^2 = 8x$

So graph is parabolic

4. The galvanometer will not show any deflection.

5. When bar becomes vertical

$\omega = \sqrt{\frac{3g}{\ell}}$ so $V = \sqrt{3g\ell}$

$B = \frac{\mu_0}{4\pi}\frac{q\sqrt{3g\ell}}{\ell^2} = \frac{\mu_0 q}{4\pi\ell}\sqrt{\frac{3g}{\ell}}$

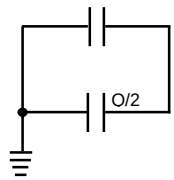
6. If $I_G = 0$ then it becomes a balanced whetstone bridge

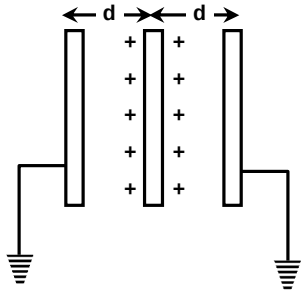
$\ell_1 = \frac{2}{3}\times 1m$

$= \frac{2}{3}\times 100 = \frac{200}{3}cm$

8. $E_1 = \frac{\sigma}{\epsilon_0} = \frac{Q}{2A\,\epsilon_0}$

$V = E_1 d = \frac{Qd}{2A\,\epsilon_0}$





9. Apply gaus theorem

$E\left(4\pi r^2\right) = \frac{q_{in}}{\epsilon_0}$

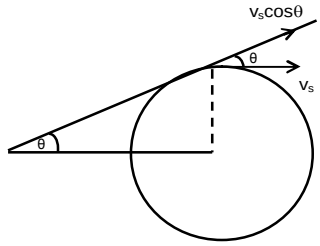
$\Rightarrow E = \frac{q}{4\pi\,\epsilon_0\, r^2} \Rightarrow E \propto \frac{1}{r^2}$

$\Rightarrow$ field depends only on r.

10. $\cos\theta = 2\sqrt{2}\dfrac{r}{\sqrt{8r^2 + r^2}}$

$\cos\theta = \dfrac{2\sqrt{2}}{3}$

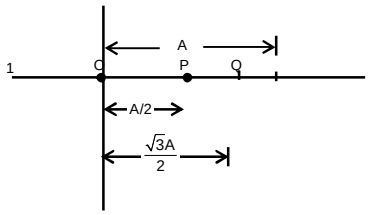
$f^1 = \dfrac{cf_0}{c + v_s\cos\theta} = \dfrac{cf_o}{c + 2\omega\dfrac{r\times 2\sqrt{2}}{3}} = \dfrac{3cf_0}{3c + 4\sqrt{2}\omega r}$



11. $\dfrac{A}{2} = A\sin\omega t_1 \Rightarrow t_1 = \dfrac{\pi}{6\omega}$

$\dfrac{\sqrt{3}}{2}A = A\sin\omega t \Rightarrow t_2 = \dfrac{\pi}{3\omega}$

$t_2 - t_1 = \dfrac{\pi}{6\omega} = \dfrac{\pi\times T}{6\times 2\pi} = T/12$



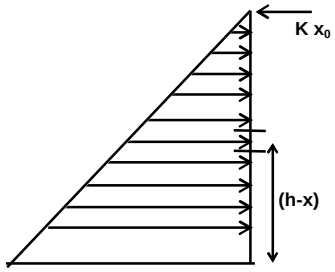
12. $kx_0h = \int_o^h (bdx)\rho gx(h-x)$

$\Rightarrow kx_0h = b\rho g\int_0^h \left(hx - x^2\right)dx = b\rho g\left[h\dfrac{h^2}{2} - \dfrac{h^3}{3}\right]$

$\Rightarrow kx_0h = b\rho g\dfrac{h^3}{6} \Rightarrow \grave{}x_0 = \dfrac{b\rho gh^2}{6k}$

$PE = \dfrac{1}{2}kx_0{}^2 = \dfrac{1}{2}k\dfrac{b^2\rho^2g^2h^4}{36k^2}$

$= \dfrac{b^2\rho^2g^2h^4}{72k}$



13. When pure rolling starts friction reduces to zero. So plank does not have any acceleration so surface is horizontal.

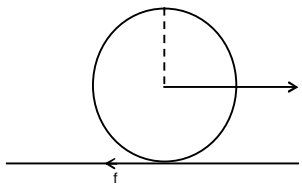
14. $F - f = ma$

$fr = \dfrac{mr^2}{2}\dfrac{a}{r}$

$\Rightarrow F = \dfrac{3}{2}ma \qquad \Rightarrow a = \dfrac{2F}{3m}$

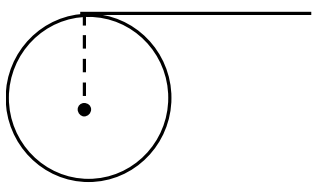
$v_{cm} = \dfrac{2Ft}{3m}$

$P = \vec{F}\cdot\vec{V} = \dfrac{2F^2}{3m}.t$



15. $I = mr^2 + \dfrac{m(2r)^2}{12} + m\left(\sqrt{2}r\right)^2$

$\Rightarrow I = 3mr^2 + \dfrac{mr^2}{3} = \dfrac{10mr^2}{3}$



16. $\Delta P = 10 - (10) \times 0.2$

$\Rightarrow 8 = 1 \times v$

$\Rightarrow v = 8 m / sec.$

17. As there does not exist any external horizontal force so X – coordinate of centre of mass will not charge

$\Rightarrow$ position of lower end is $(-\ell, 0)$

18. $\frac{1}{2} g t^2 = h \Rightarrow t = \sqrt{\frac{2h}{g}}$

$v_0 - a_0 t = 0 \Rightarrow \frac{{v_0}^2}{a_0} = \frac{2h}{g}$

$\Rightarrow h = \frac{{V_0}^2 g}{2a_0}$

19. The amplitude of the resultant wave is

$\sqrt{3^2 + 4^2 + 2 \times 3 \times 4 \times \cos 90}$

$= 5m$

20. $\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$

$\Rightarrow \frac{\Delta R}{R} \times 100 = \frac{\Delta V}{V} \times 100 + \frac{\Delta I}{I} \times 100$

$= 3\%$

21. Minimum energy transferred in the form of proton $= -13.6\left(\frac{1}{2^2} - \frac{1}{1^2}\right) = \frac{3}{4} \times 13.6 ev.$

from COE $\frac{1}{2} m v^2 = \frac{1}{2} m v_1^2 + \frac{1}{2} m v_2^2 + \frac{3}{4} \times 13.6$

$v_{min} = \sqrt{\frac{3 \times 13.6}{m}}$

$KE_{min} = \frac{1}{2} m v_{min}^2 = \frac{3}{2} \times 13.6 = 20.4 eV.$

22. $\lambda_p N_p dt = \lambda_d n_d dt$

$\Rightarrow \frac{\lambda_p}{\lambda_d} = \frac{N_d}{N_p} = \frac{1}{2}.$

23. Activity $A = \frac{A_0}{2^{\left(t/\tau \frac{1}{2}\right)}}$

$\Rightarrow \frac{A}{A_0} = \frac{1}{32}$

24. $\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{2}{f_2} = \frac{1}{15} + \frac{2\left({}^{m}\mu_{\ell} - 1\right)}{R_1}.$

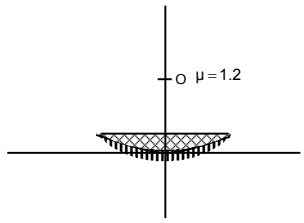
$= \frac{1}{15} + \frac{{}^{m}\mu_{\ell} - 1}{15} = \frac{{}^{m}\mu_{\ell}}{15}$

$\Rightarrow f_{eq} = \frac{15}{{}^{m}\mu_{\ell}}$

and $f_{eq} = \frac{(30 - 7.5)}{2} = \frac{22.5}{2}$

This $\frac{22.5}{2} = \frac{15}{{}^{m}\mu_1} \Rightarrow {}^{m}\mu_1 = \frac{4}{3}.$

Also ${}^{m}\mu_{\ell} = \frac{\mu_{\ell}}{\mu_m} \Rightarrow \frac{4}{3} = \frac{\mu}{1.2} \Rightarrow \mu = \frac{4 \times 1.2}{3} = 1.6$



25. For image to be coincident, either the rays should retrace or the image due to the lens should formed just at the pole of the mirror in thin case. The image formed due to lens is at 30 cm (2f) be from the lens. Thus either this image should be at centre of curvature of the convex mirror or at the pole of the mirror. Hence 6cm or 30 cm should be the separation between the lens and the mirror.

26. no of fringes shifted $= \frac{\Delta x}{\lambda} = \frac{(\mu - 1)t}{\lambda}$

$= \frac{0.48 \times 20 \times 10^{-6}}{4800 \times 10^{-10}} = 20$

27. frequency of vibration of wire will be equal to the frequency of the source hence f = 400Hz

28. $R = \frac{10}{5} = 2\Omega$

I and V are in phase $\Rightarrow x_c = x_L.$ and $Z = R$

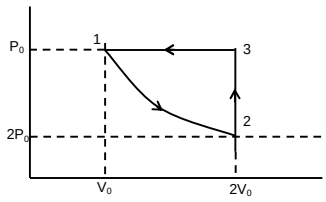
thus $I_{rms} = \frac{V_{rms}}{Z} = \frac{20}{2} = 10A.$

29. $Q_{1\to2} = \Delta U_{1\to2} + \Delta W_{1\to2}$

$\Delta U_{1\to2} = 0 \Rightarrow Q_{1\to2} = nRT_1 \ell n \frac{2V_0}{V_0}$

$Q_{1\to2} = nRT_1 \ell n2$

Also, $T_3 = 400K.$ for $1 \to 3, V \propto T \Rightarrow T_1 = \frac{T_3}{2} = 200K.$



30. Some quantity of ice will melt and the equilibrium temperature will be 0^0c

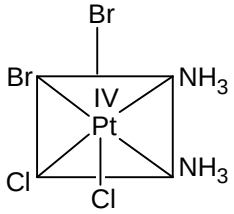
Chemistry

PART – II

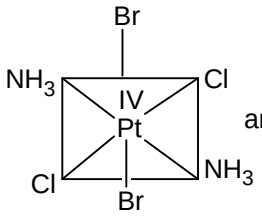
1. $SiCl_4 + 4H_2O \longrightarrow H_4SiO_4 + 4HCl$

$PCl_5 + 4H_2O \longrightarrow H_3PO_4 + 5HCl$

$BCl_3 + 3H_2O \longrightarrow B(OH)_3 + 3HCl$

2. 

(Optically active)
cis



(Optically inactive)
trans

and more stereo isomers are possible.

3. $SiH_4 + 4H_2O \longrightarrow SiO_2.2H_2O + 4H_2$

$2Al + 2NaOH + 2H_2O \longrightarrow 2NaAlO_2 + 3H_2$

$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$

$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$

6. B.O. of H_2 = I

B.O. of $H_2^+ = \frac{1}{2}$

7. With NH_3, CH_3NH_2 and $(CH_3)_2NH$, $(BH_2)(BH_2L_2)$ is formed.

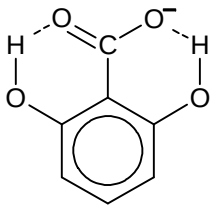
9. Ag^+, Pb^{2+} and Bi^{3+} form corresponding thiosulphate.

10. $P_4 + 3NaOH + 3H_2O \longrightarrow 3NaH_2PO_2 + PH_3$

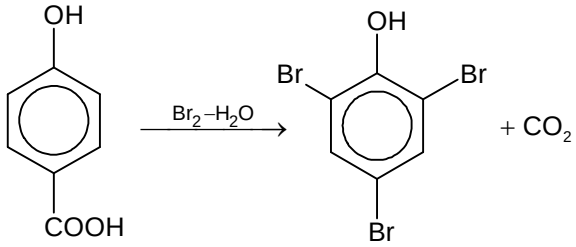
$4Na_2SO_3 \xrightarrow{\Delta} 3Na_2SO_4 + Na_2S$

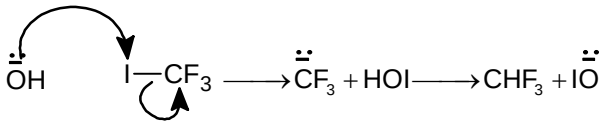
$4H_3PO_3 \xrightarrow{\Delta} 3H_3PO_4 + PH_3$

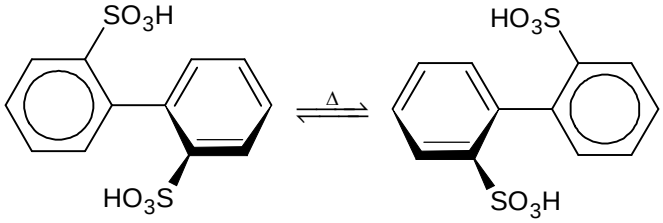
11. Due to maximum 1, 3-diaxial interaction.

12. 

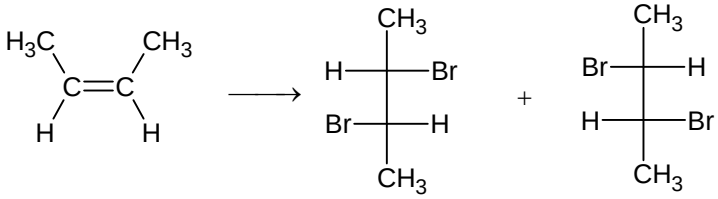
(highly stable anion).

13. 

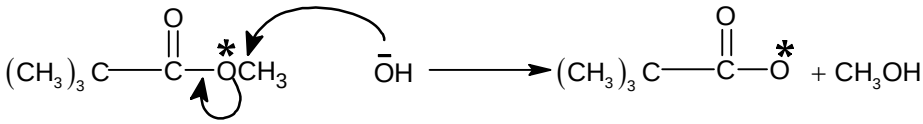
14. 

15. (A) 

(B) Due to tautomerism the product will become racemic mixture.

(C) 

16. All are stronger than carbonic acid.

17. 

20. 1, 2-addition does not yield syn-brominated product.

21. $C_V = \frac{3R}{2}$ (for monoatomic gas)

We know $q = \Delta E - W$

$$q = nC_V(\Delta T) - \left(-\frac{q}{2}\right)$$

$$n\Delta T = \frac{q}{2C_V}$$

$$\therefore C = \frac{q}{n\Delta T} = \frac{q}{q/2C_V} = 2C_V = 2 \times \frac{3R}{2} = 3R$$

22. $M + O_2 \longrightarrow \text{Metallic oxide}\left(O^{2-}\right)$

80 gm 20 gm 100 gm

No. of equivalents of M = no. of equivalents of O_2

$$\Rightarrow \frac{80}{E} = \frac{20}{8}$$

$$\therefore E = \frac{80 \times 8}{20} = 32$$

24. pH of aqueous solution of salt of weak acid and weak base is given by

$$pH = 7 + \frac{1}{2}[pK_a - pK_b]$$

25. H_2 and O_2 are produced at cathode and anode respectively.
At cathode:

$$2H_2O + 2e^- \longrightarrow H_2 + 2OH^-$$

$$2F \qquad 1\,mol$$

At anode:

$$2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$$

$$\frac{1}{2}mol \qquad 2F$$

Total volume of gases = 22.4 L H_2
+11.2 L O_2
33.6 L

26. s-orbital is spherically symmetrical.

30. Due to strong H-bond between acetone and chloroform, vapour pressure is less than that of ideal solution.

Mathematics

PART – III

1. $\frac{df(x)}{dx} > 0$

2. Last two digits should be 20, 32, 40, 52, 60, 64, 72, 76

3. $\frac{\sin B + \sin C}{2\sqrt{\sin B \cdot \sin C}} = \sin A$
$\Rightarrow$ LHS ≥ 1, RHS ≤ 1
$\Rightarrow$ sin B = sin C and sin A = 1

4. Line joining (12, 6) and $\left(2, \frac{8}{3}\right)$ will subtend a right angle at focus (a, 0)

5. $(x - 2)(x^2 + x + 1) = 0$, $x = 2$
$(x - 1)(x - 2)(y^2 + 4) = 0$, $x = 1$, $x = 2$
No triangle possible

6. Clearly $\frac{\sin A}{A} > \frac{2}{\pi}$

7. $ae = \frac{1}{2}$, $b^2 = 10$, $a^2 = \frac{41}{4}$

8. From 1 to 9, there are $9 \times 1 = 9$ digits
From 10 to 99, there are $90 \times 2 = 180$ digits and so on
So finally total digits will take 666 pages

9. $\frac{n}{n+1} < x < \frac{n+1}{n}$

10. $n^4 + 2n^3 + 3n^2 + 2n + 1 = n^2\left(n + \frac{1}{n} + 1\right)^2$

11. $\cos^{-1} x^2 = \cos^{-1} x$
$x^2 = x$, $x = 0, 1$

12. Put sin x = t

13. q = 2 and p = 3

14. Consider $f(x) = \cos x - e^{-x}$, apply Roll's theorem

15. $\int_0^{\infty} \frac{\ln x}{x^2 + t^2} dt = \frac{\pi \ln x}{2x}$

17. $\cos 60^o = \frac{29 - AC^2}{20}$

$\Rightarrow AC^2 = 19$

$\Rightarrow R = \sqrt{\frac{19}{3}}$

18. $AC = BC = \sqrt{54}$
Mid–point of AB $\Rightarrow$ D = (–1, 3, 2)
dr's of DC = (4, 2, –4)
Equation of plane is $2x + y - 2z + 3 = 0$

20. $P'(x) = \alpha(x - 1)^2(x + 1)^2$

21. Put y = 0

22. $x^n + x^{n-1} + \ldots + x + 1 = (x - r_1)(x - r_2) \ldots (x - r_n)$
Put x = 1

23. Let $\vec{b} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$

24. Total ways 4^5
Favourable = $4^5 - 4C_1 \cdot 3^5 + {}^4C_2 \cdot 2^5 - {}^4C_3 = 240$
$P(E) = \frac{240}{4^5} = \frac{15}{64}$

25. $I = \frac{1}{\sqrt{2}} \int_{-\pi/2}^{\pi/2} \frac{e^t dt}{(e^t + 1)\cos t}$

26. Lines will intersect if $\begin{vmatrix} 2-1 & 3-4 & 4-5 \\ 1 & 1 & \lambda \\ \lambda & 2 & 1 \end{vmatrix} = 0$
$\{\lambda = -1, 0\}$

27. $y = x + 2$

29. $b_1 = -\omega$ or $-\omega^2$
$b_2 = -\omega$ or $-\omega^2$
$b_3 = -\omega$ or $-\omega^2$